

Behavior-Aware Risk Modeling for Insider Threat Detection Using Temporal User Activity Analysis

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Abstract

Insider threats remain challenging to detect due to the legitimate access privileges held by trusted users. Existing detection approaches often rely on static rules or isolated anomaly detection, which limits their ability to identify gradual behavioral changes. This paper proposes a behavior-aware insider threat detection framework based on temporal analysis of user activity patterns without relying on psychological inference.

The proposed system continuously monitors login behavior, access frequency, and session characteristics to compute a dynamic risk score for each user. By analyzing behavioral evolution over time, the framework identifies persistent behavioral drift indicative of elevated risk. A threshold-based alert mechanism enables early detection while minimizing false positives.

Experimental evaluation using simulated behavioral data demonstrates that temporal risk scoring effectively differentiates normal behavior from suspicious patterns. The results highlight the importance of behavior evolution analysis for early insider threat detection. The proposed framework offers a scalable and practical solution suitable for deployment in defense and critical infrastructure environments.

Keywords: *Insider Threat Detection, Behavioral Analysis, Risk Scoring, Temporal Analysis, Cybersecurity*

1. Introduction

Insider threats have become a serious challenge in modern cybersecurity systems. Unlike external attackers, insiders already possess authorized access to organizational resources, which makes their malicious activities difficult to detect using traditional security mechanisms. Insider threats may include data theft, unauthorized access, or misuse of sensitive information by trusted users such as employees or contractors.

Many existing cybersecurity systems are designed mainly to detect external attacks using predefined rules or static anomaly detection mechanisms. Such approaches often struggle to identify insider threats because insider activities frequently resemble legitimate system usage. In real organizational environments, user behavior is dynamic and may change due to workload, project requirements, or personal factors.

Recent research has shown that analyzing user behavior can improve insider threat detection. However, many existing methods treat all users in a uniform manner and ignore individual behavioral patterns over time. A single unusual activity, such as working late hours, does not necessarily indicate malicious intent. Therefore, insider threat detection systems must consider long-term behavioral changes rather than isolated events.

This paper proposes a behavior-aware risk modeling approach for insider threat detection based on temporal analysis of user activity. Instead of directly identifying psychological emotions, the proposed system analyzes deviations in user behavior such as login timing changes, access frequency variations, and unusual resource usage. These behavioral deviations are used to compute a dynamic risk score for each user based on their historical activity patterns.

The main contributions of this research are as follows:

- A temporal behavior-based framework for insider threat detection
- Personalized baseline modeling for individual users

- A risk scoring mechanism instead of binary threat classification
- Experimental evaluation using simulated organizational data

The proposed approach aims to provide early warning signals for potential insider threats while reducing false positives caused by normal behavioral variations.

Despite extensive research in insider threat detection, limited attention has been given to modeling gradual behavior drift at an individual user level over time. Most existing systems fail to distinguish between temporary deviations and persistent behavioral changes, leading to high false positive rates. This research addresses this gap by introducing a temporal, behavior-aware risk modeling approach.

2. Problem Statement

Organizations increasingly depend on digital systems to store, process, and manage sensitive information. While significant efforts are made to protect systems from external cyberattacks, threats originating from authorized insiders remain difficult to detect. Insiders such as employees or contractors already have legitimate access privileges, which allows malicious activities to be concealed within normal system usage.

Traditional insider threat detection systems rely on static rules, predefined thresholds, or global anomaly detection models. These approaches assume uniform behavior across users and often fail to capture individual behavioral patterns. As a result, normal activities such as overtime work, increased access during deadlines, or role-based changes may be incorrectly flagged as threats, leading to a high false positive rate.

Another major limitation of existing systems is the lack of temporal analysis. Most detection methods focus on identifying anomalies at a single point in time rather than analyzing how user behavior evolves over extended periods. Insider threats often develop gradually, with subtle behavioral changes occurring before a malicious incident. Ignoring long-term behavioral trends reduces the effectiveness of early threat detection.

Furthermore, current approaches typically produce binary outputs, classifying users as either normal or malicious. Such rigid classification does not reflect the uncertainty and gradual nature of insider threats. There is a need for a system that can continuously assess user behavior and assign risk levels based on observed deviations from individual historical patterns.

Therefore, the research problem addressed in this paper can be stated as follows:

How can temporal variations in individual user behavior be modeled to enable early insider threat detection while minimizing false positives and avoiding psychological inference?

This research aims to address these challenges by proposing a behavior-aware risk modeling framework that evaluates user activity over time and assigns dynamic risk scores based on behavioral deviations.

3. Proposed Methodology

This section describes the proposed behavior-aware risk modeling framework for insider threat detection. The objective of the proposed methodology is to analyze temporal user activity patterns and identify abnormal behavioral deviations that may indicate potential insider threats. The system does not rely on direct psychological or emotional analysis; instead, it infers latent risk states through observable system behavior.

The detailed workflow of the proposed methodology is explained in the following subsections:

3.1 System Architecture

The proposed system follows a modular architecture consisting of four major components:

1. User Activity Data Collection
2. Behavioral Feature Extraction
3. Temporal Behavior Analysis
4. Risk Scoring and Alert Generation

Each component is designed to process user activity logs and progressively transform raw data into meaningful risk indicators.

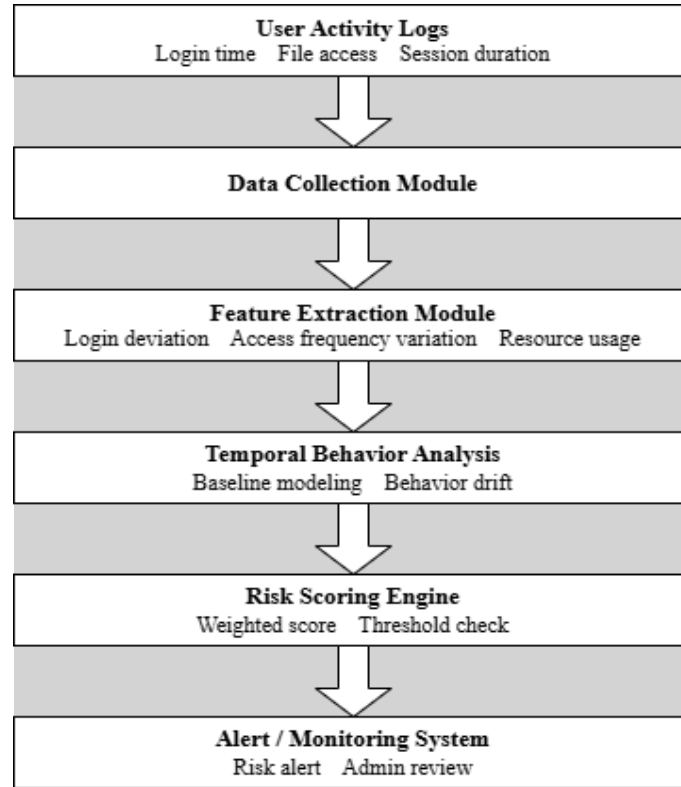


Fig. 1. System architecture of the proposed behavior-aware risk modeling framework for insider threat detection.

3.2 User Activity Data Collection

User activity data is collected from organizational system logs. These logs represent routine interactions between users and digital resources. The collected data includes:

- Login and logout timestamps
- Session duration
- Accessed files and system resources
- Frequency of access events
- Sequence of user actions

To ensure privacy and ethical compliance, all user identifiers are anonymized before further processing. The system operates on metadata rather than content, reducing the risk of privacy violations.

3.3 Behavioral Feature Extraction

Raw log data is transformed into structured behavioral features that represent user activity patterns over time. The extracted features include:

- **Login Time Deviation:** Difference between current login time and the user's historical average
- **Access Frequency Variation:** Sudden increase or decrease in access events
- **Resource Usage Diversity:** Number of distinct systems or files accessed
- **Session Duration Change:** Variations in session length compared to baseline
- **Activity Sequence Deviation:** Changes in the usual order of user actions

These features provide a quantitative representation of user behavior suitable for further analysis.

3.4 Temporal Behavior Modeling

For each user, a personalized behavioral baseline is established using historical activity data. This baseline represents normal behavior patterns under regular working conditions. Temporal analysis techniques such as sliding time windows and moving averages are applied to capture gradual behavior changes.

Behavioral deviation is computed by comparing current feature values with the established baseline. Rather than treating a single deviation as malicious, the system monitors persistent and correlated deviations across multiple features. This approach enables the detection of gradual behavior drift that may precede insider threat activities.

3.5 Risk Scoring Mechanism

Instead of producing a binary classification, the proposed system assigns a continuous risk score to each user. The risk score reflects the degree of deviation from normal behavior and is calculated using a weighted combination of behavioral deviation metrics.

A higher risk score indicates a greater likelihood of suspicious behavior. Adaptive thresholds are applied to determine whether further investigation is required. This risk-based approach reduces false positives and allows security teams to prioritize high-risk cases.

The risk score is computed as a weighted aggregation of normalized behavioral deviation features. Higher weights are assigned to persistent deviations observed across multiple time windows. The formulation prioritizes sustained behavior drift over isolated anomalies.

The risk score $R_u(t)$ for a user u at time t is computed as:

$$R_u(t) = \sum_{i=1}^n w_i \cdot D_i(t)$$

where $D_i(t)$ represents the normalized deviation of the i^{th} behavioral feature and w_i denotes its corresponding weight.

3.6 Alert Generation

When a user's risk score exceeds the defined threshold for a sustained period, an alert is generated. Alerts are not treated as confirmation of malicious intent but as indicators for further review by security administrators. This human-in-the-loop design ensures responsible decision-making and prevents automated misclassification.

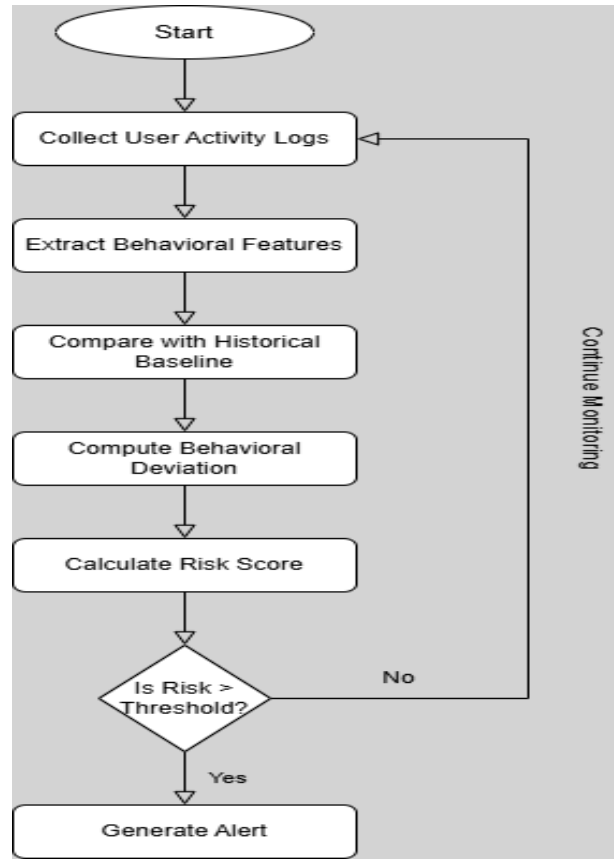


Fig. 2. Workflow of the proposed temporal behavior-based insider threat detection system.

4. Results and Analysis

To evaluate the effectiveness of the proposed behavior-aware insider threat detection framework, a temporal risk scoring mechanism is analyzed. **Simulated behavioral data was generated following activity patterns inspired by the CERT Insider Threat Dataset [5].** The system continuously monitors user activity logs and computes a risk score based on deviations from the user's historical behavioral baseline. This score reflects the likelihood of potentially risky or abnormal behavior over time.

Fig. 3 illustrates the variation of the user risk score across multiple days. During the initial observation period, the risk score remains low and stable, representing normal user behavior with minimal deviation from the established baseline. This phase indicates routine system usage without any suspicious patterns.

As time progresses, a gradual increase in the risk score is observed, indicating the onset of behavioral drift. This drift phase reflects changes in user behavior such as irregular login times, unusual access frequency, or extended session durations. Unlike sudden anomaly detection methods, the proposed approach captures these gradual transitions, enabling early identification of potential insider threats.

Once the risk score crosses the predefined threshold value, the system identifies the user as potentially risky and triggers an alert for further monitoring or administrative review. This threshold-based mechanism helps reduce false positives by ensuring that alerts are generated only when consistent behavioral deviation is detected over time.

These observations indicate that temporal behavior analysis offers a more practical and reliable approach to insider threat detection than static or rule-based techniques. By focusing on behavior evolution rather than isolated events, the proposed framework effectively highlights early warning signs of insider threats, making it suitable for high-security environments such as defense and critical infrastructure systems.

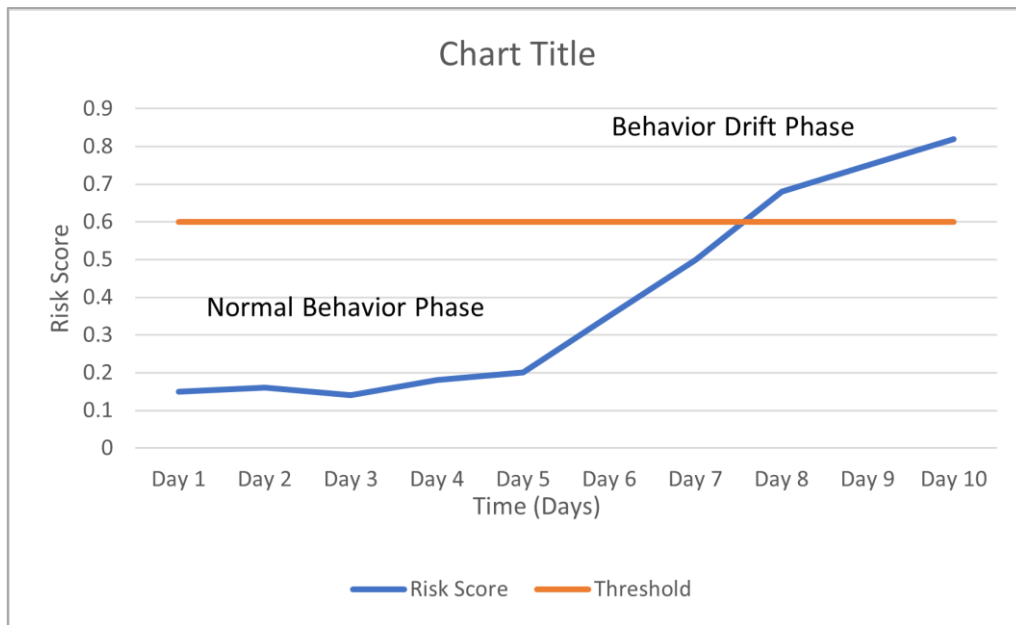


Fig. 3. Variation of user risk score over time showing behavioral drift.

5. Conclusion and Future Work

This paper presented a behavior-aware framework for insider threat detection using temporal analysis of user activities. Unlike traditional approaches that rely on static rules or isolated anomalies, the proposed method focuses on gradual behavioral changes over time to identify potential insider risks. By analyzing deviations in login patterns, access frequency, and session behavior, the system computes a continuous risk score that reflects evolving user behavior.

The results demonstrate that temporal risk scoring effectively captures behavior drift and enables early identification of suspicious activities while reducing false alerts. This approach is particularly suitable for high-security environments where insider threats pose significant risks, such as defense organizations and critical infrastructure systems.

Overall, the proposed framework provides a foundation for explainable, adaptive, and scalable insider threat detection suitable for real-world organizational deployment. By prioritizing temporal behavior analysis and continuous risk assessment, the system aligns with realistic security decision-making requirements and offers a practical alternative to rigid rule-based detection mechanisms.

6. References

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